

Supplementary Data Leakage Audit for DB-EpiFM

Audit objective

To address potential patient-level data leakage between the self-supervised pretraining corpus and downstream TUH-family evaluation datasets, we performed a patient-level overlap audit before preprocessing the pretraining corpus. The pretraining corpus consisted of TUEP and TUSZ, whereas the downstream TUH-family datasets considered in this audit were TUAB and TUEV.

Patient-level exclusion protocol

Patient identifiers were automatically extracted from the TUH directory structure and EDF file names. An exclusion list was constructed as the union of all patient identifiers appearing in TUAB and TUEV. Any EDF recording in TUEP or TUSZ whose patient identifier appeared in this exclusion list was removed before signal preprocessing. This exclusion was applied at the patient level rather than at the recording or window level.

Key audit results

Downstream patient exclusion list: 887 patient identifiers.
Actual overlapping pretraining patients removed: 101 patients.
EDF recordings removed from pretraining: 2195.
Final pretraining corpus: 162,617 non-overlapping 30-s windows (1355.1 h).
Files with unrecognized patient identifiers: 0. Signal-processing errors: 0.

Table S1. Patient-level overlap removal by pretraining source

corpus	excluded_edf_recordings	unique_excluded_pretraining_patients
TUEP	945	28
TUSZ	1250	81
TUEP+TUSZ	2195	101

Table S2. Source of patient-level overlap

overlap_with	excluded_edf_recordings	unique_excluded_pretraining_patients
TUAB	2143	89
TUEV	25	7
TUAB+TUEV	27	5

Table S3. Final pretraining corpus summary

item	value
downstream_patient_ids_exclusion_list	887
actual_excluded_pretraining_patients	101
files_excluded_by_patient_overlap	2195
files_kept_before_signal_qc	7990
files_skipped_bad_patient_id	0
files_successfully_processed	7217
files_failed_signal_processing	0

item	value
files_with_no_valid_window	773
samples_total_30s_windows	162617
samples_rejected_by_amplitude_qc	13529
samples_TUEP_30s_windows	37672
samples_TUSZ_30s_windows	124945
total_duration_hours	1355.14
total_duration_days	56.46

Released sanitized manifests

The released manifest files contain pseudonymous patient hashes and sanitized dataset-relative paths. Raw patient identifiers, local absolute paths, raw EDF files, and preprocessed LMDB databases are not redistributed.